

VANSH KAPOOR

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Technical Skills

Languages: Python, C++, JavaScript (ES6+), TypeScript, Dart, SQL

Frontend & Mobile: React.js, Flutter, Three.js, WebGL, Electron

Backend & Data: Node.js, Flask, Django, REST APIs, PostgreSQL, MongoDB, Supabase

AI/ML: TensorFlow, Scikit-learn, RAG, FAISS, Gemini API, Google ADK, Agentic AI

Cloud & DevOps: GCP (Vertex AI, Cloud Run, Firestore), CMake, Docker, Git, Vercel

Projects

High-Fidelity Synthetic Chest X-ray Generation (StyleGAN2-ADA) 2026

- Built a class-conditional StyleGAN2-ADA pipeline (8-layer Mapping Network, AdaIN, ADA, R1 regularization) synthesizing 256x256 chest X-rays to address medical-imaging data scarcity.
- Achieved FID 25.30 / KID 0.02 on 5,232 samples, enabling 97.99% VGG16 pneumonia detection accuracy; published at IEEE AIC 2026.

Aegis – Governance Runtime for Autonomous AI Agents 2026

- Co-built a policy-gated governance runtime routing every AI-agent tool call through a policy gateway (no direct agent-to-resource path), with untrusted-input screening (Model Armor), human-in-the-loop approval, crash-safe persistence, and a full audit trail; built on Gemini 3.6 Flash via Vertex AI/ADK, Cloud Run, and Pub/Sub.

Release Radar – AI-Powered Changelog Generator (Desktop App) 2026

- Built a cross-platform Electron + React app turning GitHub commit ranges into categorized changelogs; provider-agnostic layer across 4 AI APIs, OS-keychain key encryption, sandboxed renderer, shared core reused in a GitHub Action.

Experience

Architecture Lead – Cash Compass 2026

International Collab. w/ Southern Federal University, Russia *Remote*

- Leading system architecture for a cross-platform personal finance app – a React/TypeScript web dashboard and a Flutter mobile app on a shared Supabase backend – directing the Indian development team in a cross-border collaboration.
- Designed the core domain model (transactions, budgets, savings goals) and a modular drag-and-drop widget system powering a multi-theme customizable dashboard.

Open Source Contributor – Sourcemeta Core & Blaze 2026

6 merged PRs – C++ JSON Schema validation engine *Remote*

- Integrated the mimalloc allocator across ~30 C++ libraries via CMake ([Core #2708](#)), yielding 40–70% latency gains across core benchmarks; debugged a CMake export-set collision and an AddressSanitizer/allocator CI crash to land it.
- Found and fixed a soundness bug in Blaze’s canonicalizer ([#1058](#)): distributing keywords over Draft 3 type unions silently weakened `additionalProperties`, making the validator accept documents it should reject.
- Eliminated `unevaluatedProperties` annotation tracking across `$ref` branches ([#1053](#), [#1028](#)), fixed a canonicalizer crash with nested-id coverage ([#1052](#)), and replaced indirect dispatch with direct calls ([#975](#)).

Freelance Web Developer – V-Design 2025 – Present

3D Interactive Web Platform *Remote*

- Built and deployed 3 client sites (incl. [Anotherskyexplore](#), [ICCAIML](#)) with React.js, Three.js, and WebGL, generating INR 30,000 in revenue; reusable components optimized for 3D rendering.

Student Placement Coordinator – AIML Department 2025 – Present

Manipal University Jaipur *Jaipur, India*

- Coordinating recruitment processes and employer outreach for 300+ students.

Education

Manipal University Jaipur

B.Tech CSE (AI & ML)

2023 – 2027

CGPA: 7.96 (till 6th sem)

Publications

IEEE AIC 2026

High-Resolution Synthetic Augmentation for Pneumonia Detection

Co-author; presented at the 5th World Conf. on Applied Intelligence & Computing

Achievements

Smart India Hackathon 2025

National Qualifier

Hobbies

Learning Japanese

Music production & instruments

Game strategy & design

Certifications

Azure AI Fundamentals

Oracle SQL Certification

Red Hat DBMS

NPTEL – Machine Learning, DAA, Deep Learning

Relevant Coursework

Data Structures & Algorithms

DBMS, OOPs

Machine Learning, Deep Learning